

SAM10 MIDTERM PROJECT — SYSTEM PROPOSAL

Deliverable 1: 1-Page System Proposal | MediCare Direct System

Student Name: Robert Angel B. Santiago	Course & Section: SAM10 / BSIT-4-A
Student ID: 23-1360-266	Instructor: Romack L. Natividad, MIT
System Name: MediCare Direct	Institution: AUF College of Computer Studies

1. SYSTEM DESCRIPTION & BUSINESS IDEA (~120 WORDS)

MediCare Direct is a cloud-based outpatient clinic management and scheduling platform designed to streamline doctor availability tracking, patient registrations, and consultation queue management. The platform serves three main user groups: (1) **Patients** searching for clinic schedules and booking appointments online; (2) **Clinic Staff and Healthcare Providers** managing daily patient rosters and retrieving electronic health records (EHR); and (3) **System Administrators** maintaining network availability and data compliance. The major infrastructure requirements include a public-facing web interface, an isolated backend API processing engine, an encrypted database tier for clinical records, subnet segmentation, stateful firewall rules, and role-based IAM administrative controls.

2. SYSTEM REQUIREMENTS TO AWS SOLUTION MAPPING

System Requirement	Proposed AWS Service	Technical Specification & Architectural Purpose
Public Website Portal	Amazon EC2 (Web Tier)	t3.micro instance running Amazon Linux 2023 & NGINX reverse proxy for SSL/TLS termination.
Application Processing	Amazon EC2 (App Tier)	t3.small instance executing Node.js Express API scheduling logic in a private subnet.
Persistent Clinical Data	Amazon EBS (Data Tier)	100 GB gp3 SSD volume encrypted with AWS KMS (AES-256) storing PostgreSQL clinical data.
Network Isolation	Amazon VPC & Subnets	10.0.0.0/16 VPC segmented into Public (10.0.1.0/24), Private App (10.0.2.0/24), and Data (10.0.3.0/24) subnets.
Network Security	AWS Security Groups	Stateful firewalls using Security Group ID chaining (sg-WebTier -> sg-AppTier -> sg-DBTier).
Administrative Access	AWS IAM & SSM	Role-Based Access Control (RBAC) with MFA and Systems Manager Session Manager (SSH Port 22 closed).

3. NETWORK TOPOLOGY & DESIGN RATIONALE SUMMARY

- AWS Region & Network CIDR:** Deployed in **ap-southeast-1 (Singapore)** within a **10.0.0.0/16 VPC**.
- Subnet Segmentation:** Public Web Tier (10.0.1.0/24) routes traffic to the Internet Gateway (0.0.0.0/0 -> IGW). Private App (10.0.2.0/24) and Data (10.0.3.0/24) subnets contain no direct Internet routes.
- Architectural Alignment:** Aligns with the 5 Cloud Design Lenses: *Segmentation* (isolates DB from public traffic), *Least Privilege* (restricts ingress ports), *Controlled Connectivity* (enforces unidirectional flow), *Scalability* (/24 CIDRs allow 250+ hosts per tier), and *Simplicity* (eliminates unneeded managed services).